

Dual-Tree Complex Wavelet Driven Hierarchical Spatial-Frequency Fusion Learning for Robust Deepfake Detection

Rui Sun¹, Member, IEEE, Yifan Zhang, Xiaolu Yu¹, Meng Li¹, Senior Member, IEEE, and Jun Gao¹

Abstract—Deepfake detection aims to identify synthetic visual content generated by artificial models. With the improvement of generative models, high-fidelity forgeries largely weaken the explicit traces on which traditional methods rely, leaving only potential implicit clues such as subtle frequency-domain artifacts, creating an urgent need to mine robust discriminative features from either fundamental signal patterns or high-level semantic representations. Specifically, we observe that concentrating on the collaborative frequency information from different hierarchies can enhance model robustness. We propose a Hierarchical Spatial-Frequency Fusion (HSFF) module that couples multi-scale feature extraction with frequency attention to capture multi-range forgery traces, and adaptively weights frequency-domain features across scales to highlight cross-scale anomalies caused by forgeries. While HSFF enhances sensitivity to spatial-frequency inconsistencies, it does not explicitly model directional forgery patterns. To complement this, we further introduce a Collaborative Multi-Subband Enhancement (CMSE) mechanism that integrates Dual-Tree Complex Wavelet Transform (DTCWT) with pre-trained visual features, explicitly modeling interactions among six directional subbands and supplying the directional selectivity that HSFF lacks, together forming a joint spatial-frequency-direction representation that significantly improves robustness against multi-directional forgeries. The superiority of our approach is verified by extensive comparative experiments and ablation studies on multiple benchmark datasets against numerous state-of-the-art methods. In particular, our method achieves 97.2% and 89.1% cross-dataset AUROC on the challenging Celeb-DFv2 and DFDC respectively.

Index Terms—Deepfake detection, DTCWT, spatial-frequency.

I. INTRODUCTION

WITH the rapid advancement of generative models, a growing number of facial manipulation techniques have evolved, and we are witnessing a proliferation of forged images that are highly photorealistic and nearly imperceptible. For instance, deepfake enables facial substitution while preserving inherent attributes such as orientation and expression (Fig. 1).

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The authors are with the Key Laboratory of Knowledge Engineering with Big Data (Ministry of Education), School of Computer and Information, Anhui Province Key Laboratory of Industry Safety and Emergency Technology, Hefei University of Technology, Hefei 230009, China (e-mail: sunrui@hfut.edu.cn).

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Fig. 1. Examples of Deepfake from [13], which involves replacing the original faces with manipulated faces while maintaining identical facial expressions.

As a consequence, the risks of personal privacy infringement and the dissemination of disinformation have increased significantly. Therefore, developing a universal deepfake detector capable of identifying multiple existing forgery methods has become an urgent priority.

The wide variety of deepfake generators gives rise to multiple types of forgery marks, further exacerbating this challenge. These forgeries may manifest as photorealistic human faces created by generative adversarial networks (GANs) [1], [2] or as complex scenes synthesized by Diffusion [3] models. The ever-increasing diversification of forgery techniques implies that new forms of image forgeries will inevitably continue to emerge.

Diffusion models are capable of generating highly realistic and diverse synthetic images, including lifelike human faces, intricate natural scenes, and tampered content that is hard to distinguish [4]. Consequently, their rapid development has also raised serious concerns about their susceptibility to malicious misuse, such as generating deepfake media or spreading disinformation [5]. The advancement of diffusion models has made it increasingly difficult to distinguish between real and fake images, thereby imposing higher requirements on deepfake detection systems, which need to achieve generalization across different generative models. In the face of these challenges, our research aims to design a deepfake detection framework with diversity and generalization capabilities, mining robust discriminative clues from fundamental signal features, enabling it to identify any forged image even when training is confined to a single type of generative model.

Traditional deepfake detection, which is regarded as a binary classification task, utilizes deep neural networks trained to distinguish between real images and synthetic images generated by specific models, aiming to mine forgery clues from global image properties including texture anomalies [6], color distortions [7], [8], and frequency-domain artifacts [9], [10]. For instance, FrePGAN [11] mitigates frequency-level artifacts via disturbance plots and incorporates image-level irregularities in

its final iteration. These methods struggle to generalize beyond their training data due to an over-reliance on specific fingerprint features [12]. As a result, despite strong in-domain performance on specific datasets [13], [14], these methods suffer significant degradation when faced with highly realistic deepfakes from unseen generative models, this stems from forgeries lacking these artifacts evade detection, biasing the decision boundary and undermining robustness against novel generation techniques.

To address the limitations of existing methods, we instead leverage the large-scale pre-trained features of CLIP-ViT [15], which are inherently unrelated to deepfake detection. However, directly applying such general-purpose representations poses a challenge: they lack fine-grained sensitivity to forgery-specific frequency artifacts. Recently, several CLIP-based detectors—such as ForAda [16], Effort [17], and GenD [18]—have been proposed to improve cross-dataset generalization. Their common strength lies in inheriting robust semantic representations from vision-language pre-training, which mitigates overfitting to dataset-specific manipulations. Nevertheless, these methods primarily operate in the spatial domain or rely on coarse frequency statistics; they do not explicitly model directional forgery patterns or multi-scale frequency anomalies, leaving subtle artifacts in high-frequency directions underexploited. To overcome this, we mine robust discriminative clues from fundamental signal features by designing a hierarchical frequency attention mechanism that enhances frequency-domain features at the encoder output. Furthermore, capitalizing on the advantages of DTCWT [19] in directional selectivity, shift-invariance, and sub-band redundancy [20], we explicitly model interactions among subbands. This dynamically strengthens responses in critical directions while suppressing noise, thereby generating more discriminative representations of forgery traces. Experiments show that our method significantly outperforms baselines on generative deepfake benchmarks and demonstrates superior robustness and generalization in cross-technique and cross-dataset evaluations on unseen data.

Our contributions are summarized as follows:

- We design a collaborative multi-subband enhancement module based on the DTCWT to fully leverage the joint subband correlations, specifically designed for deepfake detection, and demonstrate its effectiveness.
- We propose a Hierarchical Spatial-Frequency Fusion (HSFF) module specifically for facial deepfake detection, which integrates frequency attention to extract multi-level information and further improves detection performance.
- We construct an easily implementable pipeline that not only functions effectively across different unseen datasets but also accurately identifies images generated by models trained on previously unseen datasets, confirming the robustness and adaptability of the proposed method in facial deepfake detection.

II. RELATED WORK

In this section, we summarize the advances of deepfake detection research from three perspectives: spatial deep detection,

frequency-domain detection, and generalization to unseen scenarios, and highlight the advantages of our method over existing approaches.

A. Spatial Deepfake Detection

Some studies [13] utilize images or frames extracted from video as input data to train binary models of Deepfake detection. Some works focus on specific areas of the face, such as the eyes [21] or mouth [22], to discern the real or fake face of those areas based on discontinuities in video. Ju et al. [23] incorporate both global spatial features and local informative features to develop a dual-branch model for training. Marra et al. [24] and Yu et al. [25] perform Deepfake detection by extracting the unique image fingerprints of GAN model. AltFreezing [26] leverages spatial and temporal anomalies to effectively detect face forgery. In addition, capturing differential features along the temporal dimension is also a hot topic in image-based deepfake detection. To capture the temporal evolution, Zhang et al. [27] introduce an innovative video-level 4D convolution (V4D) framework incorporating residual connections, enabling it to concurrently capture long-term relationships between snippets while preserving 3D features prior to inter-snippet interaction. STIL [28] block utilizes temporal difference within adjacent frames along two orthogonal directions to capture fine-grained temporal incoherence. Wu et al. [29] devise dynamic convolutional kernels that adaptively integrate long-range temporal data from contiguous snippets. FTCN [30] reduces the spatial convolution kernel size to 1, while maintaining the temporal convolution kernel size unchanged, and then employ a temporal transformer network to explore the long-term temporal coherence. By focusing on spatial inconsistencies such as improper fusion, texture mismatches, or tampering in facial regions, spatial detectors perform excellently in identifying specific manipulation behaviors, however, they may require more complex preprocessing and training strategies. These methods may effectively deal with visually perceptible forged content, yet they fail to detect hidden forgery marks.

B. Frequency Deepfake Detection

Since generative architectures such as GAN rely heavily on upsampling operations, it becomes meaningful to study the effect of upsampling operations on the entire image. Some studies [31], [32] take the spectrum as the representation of upsampling artifacts, and deeply study the influence of upsampling artifacts on forged images. ADD [33] develops a frequency attention distillation module and a multi-view attention distillation module to detect highly compressed false faces. Gu et al. [34] indicate that the exploitation of frequency information is coarse-grained, and to extract fine-grained forgery marks is not simple, so they perform fine-grained decomposition of RGB images to completely separate real and forged marks in the frequency space. F3Net [9] divides the frequency components and uses statistical differences in frequency between real and fake faces to detect them. FrePGAN [11] mitigates the effects of frequency-level artifacts by means of frequency-level disturbance plots, and takes the irregularity of image level into account in the final iteration. However, these methods do not

have an effective artifacts representation method that considers the generation process, and overlook the importance of global information for lossy image detection. WaveGuard [35] introduces a proactive watermarking framework that leverages dual-tree complex wavelet transform for robust watermark embedding in high-frequency sub-bands and employs structural consistency graph neural network to ensure imperceptibility, significantly enhancing deepfake detection and source tracing performance. In addition, WATCHER [36] employs a Haar wavelet-guided autoencoder to restore high-frequency textures and learns hierarchical content-texture attention maps. SFCF [37] uses stationary wavelet transform (SWT) to extract multi-scale high-frequency features and fuses them with spatial features via cross-fusion. RLGC [38] focuses on gradient-content reconstruction learning with Sobel-based frequency clues. However, these methods either treat wavelet subbands independently or lack explicit directional decomposition, thus failing to model inter-subband correlations. In contrast, our approach performs hierarchical frequency attention enhancement at the encoder output and introduces a CMSE module based on the DTCWT, which provides six directional subbands. The CMSE module explicitly models inter-subband correlations via a learnable similarity matrix and adaptive pull-closer/push-apart mechanisms, dynamically pulling closer subband pairs with abnormally low similarity and pushing apart those with abnormally high similarity. This allows our model to capture direction-aware forgery patterns that are inaccessible to prior wavelet-based methods. Together with hierarchical frequency attention, this design covers the complete frequency spectrum of forgery watermarks, enhances the perception of global structural anomalies, and improves both detection robustness.

C. Unknown Deepfake Generalization

Achieving robust generalization across diverse and evolving generative models is a core challenge in deepfake detection. Ojha et al. [39] leverages fixed, large vision-language feature spaces, circumventing the poor generalization of traditional real-vs-fake classifiers to novel generative model families. Data augmentations such as blur and JPEG can improve generalization [40]. These adaptation approaches can be categorized by their modification strategy. Effort [17] applies singular value decomposition to decompose model weights into orthogonal subspaces, freezing the principal components to retain pre-trained knowledge and fine-tuning only the residual components on forgery patterns. RSG-DA [41] enhances generalization through dynamic landmark diffusion generation and dual augmentation, but it remains susceptible to fine-grained forgery details and performance degradation under severe compression artifacts. Forensics Adapter (ForAda) [16] introduces a parallel branch explicitly trained to detect blending artifacts, enabling forgery-specific learning while preserving the original large vision-language features. Xu et al. [42] exploits pairwise learning and complementary multi-channel color space information. Compared to these works, our method deeply integrate frequency-domain analysis within the adaptation process. While existing approaches largely operate in spatial or parametric domains,

our method explicitly models hierarchical frequency interactions and inter-subband correlations—features typically underutilized in prior detectors—thereby preserving inherent generalization while significantly enhancing sensitivity to frequency domain forgery artifacts.

III. APPROACH

Our approach is driven by a key insight: while CLIP-ViT offers strong generalization as a frozen feature extractor, it lacks fine-grained frequency awareness essential for deepfake detection. To bridge this gap, we design two complementary modules that operate on its classification token representations. As shown in Fig. 2, we adopt the CLIP-ViT-L/14 image encoder (producing $16 \times 16 + 1$ tokens of dimension 1024) and freeze all its weights to retain cross-dataset robustness. Only the classification token is used, discarding patch tokens to reduce noise. We then enhance this representation via: a) HSFF, which applies hierarchical frequency attention to capture multi-scale forgery patterns, and b) CMSE, which models directional subband dependencies using DTCWT to emphasize critical forgery directions while suppressing irrelevant variations. Finally, the refined features are classified by a trainable linear layer ($W \in \mathbb{R}^{1024 \times 2}, b \in \mathbb{R}^2$) with cross-entropy loss.

A. Hierarchical Spatial-Frequency Fusion

We adopt the ViT-L/14 architecture pretrained on the CLIP model as the encoder, and integrates HSFF module at its end to enhance the effectiveness of high-frequency feature extraction.

The HSFF module establishes a hierarchical feature processing pipeline that combines frequency-domain attention with spatial feature preservation. Given an input feature map $X \in \mathbb{R}^{B \times C \times H \times W}$, which is first passed through a downsampling convolutional layer (*down_conv*) to reduce the spatial dimensions, and then through a convolutional layer (*Conv₁*) with a kernel size 1×1 to extract core features, denoted as:

$$F_1 = \text{Conv}_1(\text{down_conv}(X)) \quad (1)$$

Then it is fused with spatial information and high-frequency information through a multi-level Fourier Attention Block, denoted as:

$$F_i = \mathcal{H}_i(\text{Conv}_{2i-1}(F_{i-1}) + X), \quad i = 2, \dots, 5 \quad (2)$$

where $\mathcal{H}_i$ denotes the i -th Fourier Attention Block, which we will introduce below. The Conv_{2i-1} represents the convolution with a kernel size of $2i - 1$, where $i = 2, \dots, 5$, enabling simultaneous capture of both local high-frequency details with (3×3) and global low-frequency information with (9×9) . In addition, a gradual transition is constructed with intermediate sizes $(5 \times 5), (7 \times 7)$.

Finally, the features at all levels are combined with the original features processed by Global Average Pooling through a fusion convolution $\Gamma(\cdot)$, denoted as:

$$F_{out} = \Gamma([F_1, F_2, F_3, F_4, F_5, \text{GAP}(X)]), \quad (3)$$

where $[\cdot, \cdot]$ indicates concatenation, and $\Gamma(\cdot)$ represents the fusion convolution. Within the HSFF module, we construct a

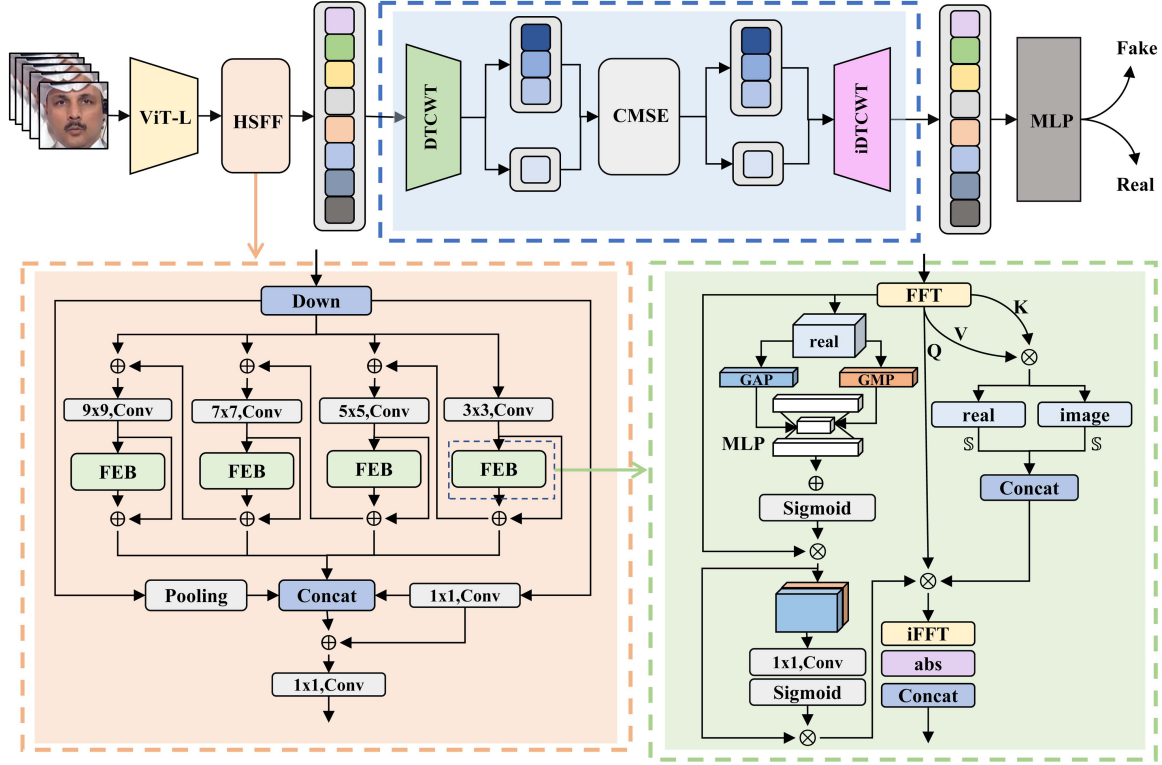


Fig. 2. The overall framework of our method. In the figure, CMSE stands for Collaborative Multi-Subband Enhancement, HSFF stands for Hierarchical Spatial-Frequency Fusion, FEB stands for Fourier Enhancement Block. $\oplus$, $\otimes$ denote element-wise addition and multiplication, respectively. $\mathbb{S}$ represents the Softmax function.

Fourier Enhancement Block (FEB) that sequentially integrates a Fourier Attention Block and a Channel Attention Block to jointly model cross-scale frequency attention and channel-wise recalibration.

1) *Fourier Attention Block*: The Fourier Attention Block mentioned above is designed to generate queries Q , keys K , and values V . For each processing stage i , we first transform the spatial features to the frequency domain:

$$Q_f^i, K_f^i, V_f^i = \mathcal{F}(F_i) \in \mathbb{C}^{B \times C \times H \times W}, \quad (4)$$

where $\mathcal{F}(\cdot)$ denotes the 2D Fast Fourier Transform (FFT). After reshaping, the queries $\tilde{Q}$ and keys $\tilde{K}$ generate the transpose attention map A_f by Hadamard product $\odot$ and then a complex normalization, denoted as:

$$Z = \tilde{Q} \odot \tilde{K}, \quad (5)$$

$$A_f = \text{Softmax}(Z^{\text{re}}) + j\text{Softmax}(Z^{\text{im}}), \quad (6)$$

where A_f is the new complex number combined from the real parts (Z^{re}) and imaginary parts (Z^{im}) of the intermediate variable Z after respective activation.

Subsequently, the attention map A_f performs a dot product with V for weighted optimization, and the result is converted back to the original domain via the Inverse Fast Fourier Transform (IFFT).

$$O_f^i = \|\mathcal{F}^{-1}(A_f^i V_f^i)\| \quad (7)$$

2) *Channel Attention Block*: To adaptively enhance the spatial features corresponding to key frequency information, we define a parallel two-dimensional Channel Attention Branch on the frequency features.

This branch adaptively recalibrates channel-wise features of the real part of the frequency-domain representation by introducing a dual-path global pooling attention mechanism, which integrates both Global Average Pooling (GAP) and Global Max Pooling (GMP) to capture complementary statistical cues across channels. GAP captures holistic spectral biases commonly introduced by generative models—such as unnatural smoothing or consistent high-frequency attenuation across frames—which manifest as shifted energy distributions in the frequency domain, and GMP highlights localized, salient anomalies, such as boundary inconsistencies, blending artifacts, or residual grid-like patterns near manipulated regions, which often correspond to strong but sparse frequency responses.

Let $T_f^i = \mathcal{F}(F_i)$ denote the Fourier transform of the input feature map F_i , and let $T_f^{i,\text{re}} = \text{Re}(T_f^i)$ represent its real component. The proposed hierarchical frequency attention enhancement mechanism is formulated as follows:

First, dual-pooling based attention weights are computed from the real component:

$$A_{\text{init}} = \sigma \left(g_{\text{amp}}(\text{GAP}(T_f^{i,\text{re}})) + g_{\text{amp}}(\text{GMP}(T_f^{i,\text{re}})) \right) \odot T_f^{i,\text{re}}, \quad (8)$$

where $\text{GAP}(\cdot)$ and $\text{GMP}(\cdot)$ denote global average pooling and global max pooling operations, $g_{\text{amp}}(\cdot)$ is a learnable gating function implemented as a multi-layer perceptron (MLP), and $\sigma(\cdot)$ is the sigmoid activation function.

Next, channel-wise statistics are aggregated via max and average pooling along the channel dimension:

$$M_{\text{cat}} = \text{concat}(\max_c(A_{\text{init}}), \text{avg}_c(A_{\text{init}})) \quad (9)$$

where $\max_c(\cdot)$ and $\text{avg}_c(\cdot)$ denote max and average pooling along the channel dimension. Finally, a 1×1 convolutional layer $\text{Conv}_1(\cdot)$ followed by a sigmoid activation is applied to generate the refined complex attention mask, and the enhanced complex output is obtained by element-wise multiplication. The final output is obtained by applying the inverse Fourier transform and taking the modulus:

$$O_c^i = \left\| \mathcal{F}^{-1} \left(\sigma \left(\text{Conv}_1(M_{\text{cat}}) \right) \odot T_f^i \right) \right\|. \quad (10)$$

By jointly exploiting both average and maximal channel responses, this dual-path design enriches the representational capacity of channel attention in the frequency domain, offering a more robust and discriminative feature recalibration.

The hybrid features are formed by fusing O_f^i and O_c^i through a 1×1 convolution and residual connection with the pre-FFT features, enhancing discriminative forged artifact information through joint spectral-spatial representation learning, denoted as:

$$F_{i+1} = \text{Conv}_1(O_f^i, O_c^i) + F_i \quad (11)$$

B. Collaborative Multi-Subband Enhancement

In spectral signal processing, low-frequency components encode essential structural patterns while high-frequency components capture localized details. Conventional frequency transforms such as the DCT provide only global frequency magnitude information and lack directional selectivity, making them insensitive to orientation-specific forgery traces. In contrast, the DTCWT decomposes the input into six directional high-frequency subbands, enabling finer delineation of image structures. Moreover, DTCWT is approximately shift-invariant due to its dual-tree structure, whereas DCT is sensitive to small spatial shifts, which is critical for detecting forged faces that may undergo geometric transformations. Compared to DWT, DTCWT decomposes the input into six directional high-frequency subbands, corresponding to angles of 15° , 45° , 75° , 105° , 135° and 165° , enabling finer delineation of image. As shown in Fig. 3, the DTCWT employs a circular layout to intuitively illustrate the directional and hierarchical structure of its two-level decomposition. The first-level decomposition divides the image into six high-frequency subbands ($y_{1,d}^h$, $d = 1 \sim 6$) and a central low-frequency sub-band (y_l), and the second-level decomposition further refines the first-level low-frequency sub-band, generating six new high-frequency subbands ($y_{2,d}^h$) and a low-frequency sub-band, with the spatial resolution of each sub-band halved. Black arrows indicate the first-level high-frequency sub-bands,

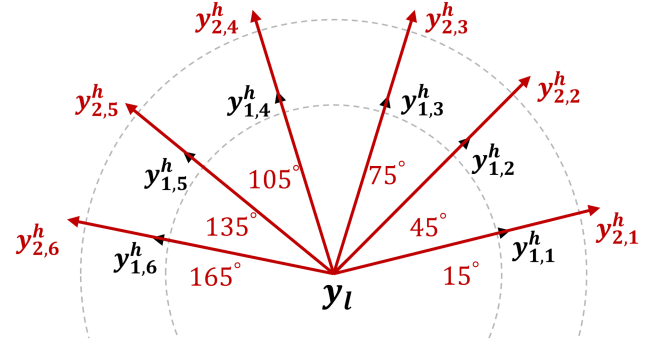


Fig. 3. Two-level DTCWT decomposition.

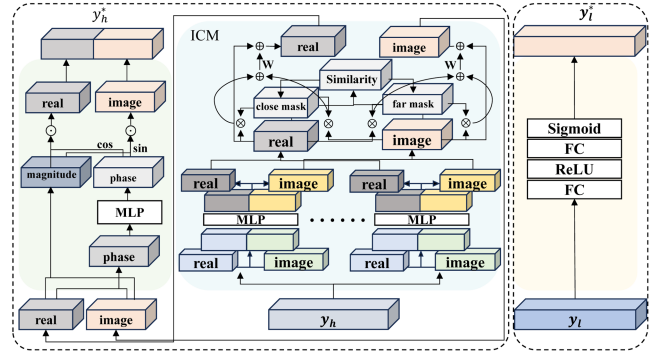


Fig. 4. The structure of the CMSE module. We enhance low-frequency components via SE, and decompose high-frequency components by DTCWT. For the six directional subbands, we perform real/imaginary separation, individual MLP enhancement, inter-subband collaboration (ICM), and phase enhancement while preserving magnitude.

red arrows indicate the second-level high-frequency sub-bands, and the directions of the sub-bands are strictly aligned across both levels.

DTCWT better captures fine-grained structural details while maintaining robustness against distortions. Crucially, DTCWT naturally forms joint subband correlations during decomposition—high-frequency subbands across different directions and scales are not entirely independent but exhibit structured information overlap and mutual reinforcement [35]. Although this redundancy can enhance robustness in traditional applications, its potential collaborative mechanisms remain underexplored. Most existing methods still treat subbands as independent processing units, failing to explicitly model and leverage the complementary and reinforcing relationships among subbands.

Our proposed CMSE achieves efficient deepfake detection through frequency-adaptive feature learning. It explicitly models and enhances interactions across directional subbands, enabling more discriminative frequency-domain representations for detecting forged patterns.

Specifically, as shown in Fig. 4, we exploit the critically valuable low-frequency information y_l in dual-tree complex wavelet decomposition and enhance it via one-dimensional

squeeze-and-excitation mechanisms,

$$\hat{y}_l = \sigma(W_4 \delta(W_3 y_l)) \odot y_l, \quad (12)$$

where $W_3 \in \mathbb{R}^{L/r \times L}$, $W_4 \in \mathbb{R}^{L \times L/r}$ represent the two learnable weights, r represents reduction ratio, δ represents ReLU, σ represents Sigmoid and $\odot$ represents the Hadamard product.

High-frequency information $y_{j,d}^h \in \mathbb{C}^{B \times 1 \times 6 \times H_j \times W_j \times 2}$, where H_j and W_j denote the spatial dimensions of the corresponding scale. For each scale j , the complex high-frequency coefficients are separated into their real parts $R_d^{(j)}$, and imaginary parts $I_d^{(j)}$. For the d -th direction, after concatenating its real and imaginary components along the channel dimension to form the combined feature, a direction-specific attention network using an MLP layer is applied:

$$C_d^{(j)} = \text{MLP} \left(\text{concat} \left[R_d^{(j)}, I_d^{(j)} \right] \right) \quad (13)$$

The enhanced features $C_d^{(j)}$ are split back into new real and imaginary components.

After independently enhancing all six directional sub-bands, the enhanced real and imaginary components from each direction are stacked separately along the directional dimension. Then we flat the spatial dimensions, and apply L2 normalization along the last dimension, denoted as $\hat{R}^{(j)}$ and $\hat{I}^{(j)}$:

$$\hat{R}^{(j)} = \frac{R_{fl}^{(j)}}{\|R_{fl}^{(j)}\|_2 + \epsilon}, \quad \hat{I}^{(j)} = \frac{I_{fl}^{(j)}}{\|I_{fl}^{(j)}\|_2 + \epsilon}, \quad (14)$$

where $\epsilon = 10^{-8}$ is a small constant for numerical stability and $R_{fl}^{(j)}$ and $I_{fl}^{(j)}$ denote the flattened real and imaginary components, respectively. The inter-subband similarity matrix is then computed via matrix multiplication:

$$S^{(j)} = \hat{R}^{(j)} \left(\hat{R}^{(j)} \right)^\top + \hat{I}^{(j)} \left(\hat{I}^{(j)} \right)^\top. \quad (15)$$

This similarity matrix integrates both real and imaginary components to quantify the correlations among the six directional sub-bands.

Given the inter-subband similarity matrix $S^{(j)} \in \mathbb{R}^{B \times 6 \times 6}$, the adaptive enhancement mechanism computes the average similarity and generates pull-closer and push-apart enhancement matrices. First, define the mask matrix $M = J_{6 \times 6} - I_6 \in \mathbb{R}^{1 \times 6 \times 6}$, where I_6 is the identity matrix, to exclude diagonal elements. The average similarity for each batch sample is calculated as:

$$\bar{S}^{(j)} = \frac{\sum_{i,k} S^{(j)}[b, i, k] \cdot M[0, i, k]}{\sum_{i,k} M[0, i, k] + \epsilon} \in \mathbb{R}^B \quad (16)$$

where $\epsilon = 10^{-8}$. Based on the average similarity, define the pull-closer threshold $\tau_c^{(j)} = \bar{S}^{(j)} - \delta$ and the push-apart threshold $\tau_f^{(j)} = \bar{S}^{(j)} + \delta$, where the δ denotes the threshold offset.

The pull-closer enhancement matrix $\Delta_c^{(j)}$ enhances subband pairs (i, k) with excessively low similarity (satisfying $0 < S^{(j)}[b, i, k] < \tau_c^{(j)}[b]$):

$$\Delta_c^{(j)} = \alpha \cdot \left(\tau_c^{(j)} \otimes J_{6 \times 6} - S^{(j)} \right) \odot M_c^{(j)} \quad (17)$$

where $M_c^{(j)}[b, i, k] = \mathbb{I}(0 < S^{(j)}[b, i, k] < \tau_c^{(j)}[b])$ is the pull-closer mask, α is the learnable pull-closer coefficient and $\otimes$ denotes the outer product.

The push-apart enhancement matrix $\Delta_f^{(j)}$ suppresses subband pairs with excessively high similarity (satisfying $S^{(j)}[b, i, k] > \tau_f^{(j)}[b]$):

$$\Delta_f^{(j)} = \beta \cdot \left(S^{(j)} - \tau_f^{(j)} \otimes J_{6 \times 6} \right) \odot M_f^{(j)} \quad (18)$$

where $M_f^{(j)}[b, i, k] = \mathbb{I}(S^{(j)}[b, i, k] > \tau_f^{(j)}[b])$ is the push-apart mask and β is the learnable push-apart coefficient.

This adaptive mechanism dynamically adjusts the enhancement strength based on the deviation of inter-subband similarity from the average level, pulling closer subband pairs with excessively low similarity and appropriately separating those with excessively high similarity, thereby optimizing the collaborative representation among subbands.

Let $R_{fl}, I_{fl} \in \mathbb{R}^{B \times 6 \times N}$ (where $N = H \times W$) represent the flattened real and imaginary components. The enhanced features are obtained through weighted combination of the original and collaboratively transformed components:

$$\begin{aligned} \hat{R}_{fl} &= R_{fl} + W \odot (\Delta_c^{(j)} \otimes R_{fl} - \Delta_f^{(j)} \otimes R_{fl}) \\ \hat{I}_{fl} &= I_{fl} + W \odot (\Delta_c^{(j)} \otimes I_{fl} - \Delta_f^{(j)} \otimes I_{fl}) \end{aligned} \quad (19)$$

where $W \in \mathbb{R}^{B \times 6 \times 1}$ is a learnable weighting factor applied via broadcasted element-wise multiplication. The enhanced features $\hat{R}_{fl}, \hat{I}_{fl}$ are then reshaped back to their original spatial dimensions to get $\bar{R}, \bar{I}$.

This formulation enables adaptive modulation of cross-subband interactions while preserving the original feature structure. Then, get the magnitude and phase:

$$\begin{aligned} A^{(j)} &= \sqrt{(\bar{R})^2 + (\bar{I})^2 + \epsilon} \in \mathbb{R}^{B \times 1 \times 6 \times H_j \times W_j} \\ \Phi^{(j)} &= \arctan 2(\bar{I}, \bar{R}) \in \mathbb{R}^{B \times 1 \times 6 \times H_j \times W_j} \end{aligned} \quad (20)$$

where $\epsilon = 10^{-8}$. The phase tensor is reshaped to $\Phi_r^{(j)} \in \mathbb{R}^{B \times 6 \times H_j \times W_j}$ and then nonlinearly transformed by the phase enhancement network using an MLP layer, which is implemented as a lightweight convolutional network with two layers: a 3×3 convolution projecting the 6-channel phase input to 12 channels, followed by BatchNorm, ReLU, and another 3×3 convolution projecting back to 6 channels. This design performs a non-linear transformation of the 6-dimensional directional phase features at each spatial location, leveraging local context (kernel size 3) to preserve phase continuity:

$$\tilde{\Phi}^{(j)} = \text{MLP}(\Phi_r^{(j)}) \in \mathbb{R}^{B \times 6 \times H_j \times W_j}. \quad (21)$$

The enhanced phase is restored to $\tilde{\Phi}_u^{(j)} \in \mathbb{R}^{B \times 1 \times 6 \times H_j \times W_j}$.

The complex signal is then reconstructed using the original magnitude $A^{(j)}$ and the enhanced phase $\tilde{\Phi}_u^{(j)}$:

$$\begin{aligned} \tilde{R}^{(j)} &= A^{(j)} \odot \cos(\tilde{\Phi}_u^{(j)}) \\ \tilde{I}^{(j)} &= A^{(j)} \odot \sin(\tilde{\Phi}_u^{(j)}) \end{aligned} \quad (22)$$

Finally, the real and imaginary parts are concatenated along the last dimension to form the enhanced complex high-frequency coefficients:

$$\hat{y}_h^{(j)} = [\tilde{R}^{(j)}, \tilde{I}^{(j)}] \in \mathbb{R}^{B \times 1 \times 6 \times H_j \times W_j \times 2} \quad (23)$$

This phase enhancement mechanism optimizes the phase information through nonlinear transformation without altering the magnitude, thereby improving the discriminative power and stability of the complex features.

Subsequently, the IDTCWT reconstructs the enhanced low-frequency feature $\hat{y}_l$ and high-frequency features $\hat{y}_h^{(j)}$ back to the spatial domain, and the reconstructed features are fed into the classifier to yield the classification scores.

IV. EXPERIMENTS

In this section, we first present the experimental settings, including the evaluated datasets, evaluation metrics, implementation details, and baselines. We then conduct cross-dataset comparison experiments. Furthermore, we perform ablation studies. Finally, we visualize the inter-frame collaboration mechanism and analyze the robustness of the model.

A. Experiments Settings

1) *Datasets*: We evaluate our method on both traditional deepfake detection datasets including FaceForensics++ (FF++) [13], DeepFakeDetection (DFD), Celeb-DFv2 [43], DFDC [44], Face Forensics in the Wild (FFIW) [45], Diffusion-Face [46] and generative deepfake detection datasets DF40 [47].

FF++[13] is a standardized benchmark for evaluating deep face forgery detection methods. It consists of 1,000 real videos extracted from YouTube and uses four manipulation techniques to generate corresponding fake videos, i.e., DeepFakes (DF), Face2Face (F2F), FaceSwap (FS), and NeuralTextures (NT), with two different compression rates including c23 and c40. Considering storage and bandwidth limitations, we utilized the c23 version. In addition, the DeepFakeDetection (DFD) dataset is also included in this version. DFD was released by Google Research in 2019 and contains 3,000 real videos and 3,000 fake videos generated by a face-swapping model.

Celeb-DFv2 [43] is a large-scale dataset for the development and evaluation of deepfake detection algorithms in deepfake detection. It includes 590 real videos and 5,639 high-quality deepfake videos from YouTube, totaling more than two million frames. The fake videos are generated through an improved synthesis method, thus exhibiting fewer significant visual artifacts compared to available datasets previously.

DFDC [44] dataset is a preview dataset and comprises approximately 5,000 videos including 4,119 fake videos generated by two synthesis methods and 1,131 real videos from 66 paid actors, many of which have been subjected to unknown manipulation methods. The presence of partially forged faces in these videos poses a significant challenge, rendering even the most advanced detectors less effective in accurately identifying manipulations within this dataset.

FFIW [45] comprises 10,000 high-quality forged videos collected from YouTube, with an average of three faces per frame.

Each video contains both real and fake faces, making it more representative of complex real-world scenarios. The manipulation process is fully automated, controlled by a domain-adversarial quality assessment network, which ensures high scalability and low human labor cost.

DiffusionFace [46] is the first diffusion-based face forgery dataset, specifically designed to evaluate deepfake detectors on high-fidelity synthetic faces generated by diffusion models. It is built upon 30,000 high-resolution real face images from the Multi-Modal-CelebA-HQ dataset, each accompanied by semantic masks, attribute labels, and descriptive text captions.

DF40 [47] is a comprehensive deepfake detection benchmark designed to address generalization challenges. It integrates 40 diverse generation methods spanning four categories: face swapping, reenactment, entire face synthesis, and face editing. It includes over 0.1 M video clips and more than 1 M images, featuring both classical techniques and state-of-the-art models such as diffusion models and GANs.

2) *Evaluation Metrics*: Following the evaluation metrics in previous works, we report video-level the area under the receiver operating characteristic curve (AUROC) for a fair comparison.

AUROC denotes the area under the Receiver Operating Characteristic (ROC) curve. The ROC curve is a two-dimensional graph with the False Positive Rate (FPR) on the x-axis and the True Positive Rate (TPR) on the y-axis. The AUROC value reflects the model's comprehensive performance across various thresholds. The formula for AUROC can be expressed as,

$$\text{AUROC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(x)) dx. \quad (24)$$

3) *Implementation Details*: We uniformly sample frames from each video, select dlib [60] as the face detector for each dataset, and then align the facial area of faces using predicted keypoints as model input. For each video, we use 32 frames to train and test all frame-based and video-based models. We employ the visual part of CLIP-ViT-L/14 as the main backbone network and adopt the PEFT [18], which fine-tunes only a small fraction of the model's parameters while keeping the majority of parameters fixed, thereby introducing a limited number of learnable degrees of freedom at a low computational cost. The batch size is set to 128 samples. For the optimization of model parameters, we utilize the Adam [61] optimizer without weight decay. The learning rate is scheduled using cosine decay. The initial learning rate is $8e-5$, which decays to $5e-5$ after 50 training epochs. The δ is set to 0.2. We use various data augmentation methods implemented in the torchvision library to expand the training set, including random horizontal flipping, random affine transformation, Gaussian blur, color jitter, and JPEG compression.

4) *Baselines*: Following the established cross-dataset generalization protocol [62], we compare our method with state-of-the-art detectors that exhibit strong cross-dataset performance, including spatial detectors (e.g., LipForensics [22], FTCN [30]), frequency-based detectors, and recent approaches (e.g., Effort [17], UGNN [55], P&P [59], Uniforensics [57], ForAda [16], GenD [18]). All methods are trained on FF++ and

TABLE I
GENERALIZATION OF MODELS TRAINED ON THE FF++ DATASET TO UNSEEN DATASETS AND FORGERY METHODS. REPORTED VALUES ARE VIDEO-LEVEL AUROC. THE BEST RESULTS ARE MARKED IN BOLD.

Methods	Publication	Year	Backbone	Celeb-DFv2	DFD	DFDC	FFIW	DiffusionFace
LipForensics [22]	CVPR	2021	ResNet-18	0.824	-	0.735	-	0.483
FTCN [30]	ICCV	2021	3DResNet-50	0.869	-	0.740	0.745	-
RealForensics [48]	CVPR	2022	ModifiedCSN	0.869	-	0.759	-	0.511
SBI [49]	CVPR	2022	EFNB4	0.932	0.827	0.724	0.848	0.494
AUNet [50]	CVPR	2023	Xception+ART	0.928	0.992	0.738	0.814	-
StyleDFD [51]	CVPR	2024	3DResNet-50	0.890	0.961	-	-	0.547
LSDA [52]	CVPR	2024	EFNB4	0.911	-	0.770	0.724	-
AltFreezing [53]	CVPR	2024	3DResNet-50	0.895	0.985	0.994	-	0.560
RLGC [38]	TCE	2024	EfficientNet-B4	0.912	0.943	0.865	-	0.579
WATCHER [36]	IJCV	2024	EfficientNet-B4	0.884	0.937	0.840	-	0.492
DiffusionFake [54]	NeurIPS	2024	EfficientNet-B4	0.832	0.917	0.774	-	0.633
TALL++ [55]	IJCV	2024	Swin-B	0.919	-	0.785	-	-
UGNN [56]	TIFS	2024	Swin-B	0.962	0.968	0.877	-	-
CLIPping [57]	ICMR	2024	CLIPViT-Large	-	0.866	0.719	-	-
UniForensics [58]	TDSC	2025	UniFormerV2	0.953	-	0.772	-	-
UDD [59]	AAAI	2025	CLIPViT-B/16	0.931	0.955	0.812	-	-
Effort [17]	ICML	2025	CLIPViT-L/14	0.956	0.965	0.843	0.921	0.628
P&P [60]	CVPR	2025	CLIPViT-L/14	0.947	0.965	0.843	0.921	-
ForAda [16]	CVPR	2025	CLIPViT-L/14	0.957	0.972	0.872	-	-
GenD [18]	arXiv	2025	CLIPViT-L/14	0.966	0.980	0.872	0.915	-
Ours	-	-	CLIPViT-L/14	0.972	0.983	0.891	0.928	0.643

evaluated on Celeb-DFv2, DFD, DFDC, FFIW, and Diffusion-Face under the same experimental setting, ensuring a fair and comprehensive performance comparison.

B. Cross-Dataset Comparisons

1) *Traditional Deepfake Datasets*: We evaluate the generalization of our model following the standard protocol: training on FF++ and testing on unseen datasets. The video-level AUROC results are presented in Table I.

Our method achieves state-of-the-art performance across all four benchmarks, with AUROCs of 0.972 on Celeb-DFv2, 0.983 on DFD, 0.891 on DFDC, 0.922 on FFIW, and 0.643 on DiffusionFace. This represents consistent improvement over recent strong baselines.

The superior and stable performance across diverse forgery methods and video qualities underscores the strong generalization capability of our framework. We attribute this primarily to the CMSE, which learns to adaptively enhance discriminative frequency representations that are less reliant on dataset-specific artifacts.

In addition, we also conduct the cross-dataset comparisons within FF++, as shown in Table II. Our method outperforms recent CLIP-based approaches such as ForAda, Effort, and GenD on three of the four forgery categories—F2F, FS, and NT—and achieves the highest average AUROC, which demonstrates its enhanced sensitivity to subtle texture-based manipulations. Although ForAda performs marginally better on the DF subset, our method delivers more balanced and robust performance across

TABLE II
VIDEO-LEVEL TEST AUROC ON CROSS-DATASET WITHIN THE FF++

Methods	DF	F2F	FS	NT	Avg.
ForAda [16]	0.997	0.970	0.986	0.919	0.968
Effort [17]	0.994	0.932	0.984	0.846	0.939
GenD [18]	0.995	0.981	0.987	0.955	0.980
Ours	0.996	0.985	0.992	0.961	0.984

all forgery types, highlighting the effectiveness of the proposed hierarchical frequency attention and subband collaborative enhancement in capturing diverse forgery traces.

2) *Generation Deepfake Datasets*: To further assess generalization on generative deepfakes, we compare our method with state-of-the-art detectors on six face-swapping forgeries from the DF40 [47] dataset. As shown in Table III, our method achieves the highest average AUROC of 0.891, outperforming all competitors. It obtains the best performance on four forgery categories—uniface, e4s, facedancer, and inswap—and remains highly competitive on the remaining two. For instance, on the challenging e4s forgery, our model reaches an AUROC of 0.889, surpassing the second-best method FSG by 1.4 percentage points. These results confirm that our approach generalizes effectively not only across traditional deepfakes but also against diverse generative face-swapping techniques, underscoring the capability of the proposed frequency-aware modules in capturing subtle artifacts produced by modern generative models.

TABLE III
COMPARISON WITH STATE-OF-THE-ART DETECTION METHODS ON SIX FORGERY CATEGORIES IN DF40 [47]

Method	uniface	simswap	facedancer	stargan	sd-2.1	midjourney6	Avg.
RECCE [64]	0.842	0.730	0.783	0.782	0.651	0.668	0.743
SBI [49]	0.644	0.568	0.447	0.743	0.598	0.612	0.602
CORE [65]	0.817	0.693	0.717	0.805	0.673	0.689	0.732
IID [66]	0.795	0.640	0.790	0.796	0.654	0.670	0.724
UCF [67]	0.787	0.649	0.800	0.812	0.682	0.695	0.738
LSDA [52]	0.854	0.727	0.759	0.834	0.701	0.718	0.766
CDFA [14]	0.765	0.761	0.754	0.820	0.720	0.761	0.764
ProDet [68]	0.845	0.778	0.736	0.845	0.715	0.732	0.775
FSG [69]	0.918	0.910	0.830	0.872	0.753	0.769	0.842
Ours	0.929	0.905	0.853	0.907	0.796	0.812	0.867

TABLE IV
ABLATION STUDY FOR PROPOSED MODULES

Methods	Celeb-DFv2	DFD	DFDC	FFIW
O	0.966	0.978	0.869	0.907
$O + H$	0.969	0.980	0.878	0.915
$O + 7 \times 7$	0.967	0.979	0.873	0.911
$O + 9 \times 9$	0.966	0.978	0.871	0.909
w/o 3×3	0.967	0.979	0.875	0.912
w/o 5×5	0.968	0.979	0.876	0.913
w/o 7×7	0.967	0.979	0.875	0.912
w/o 9×9	0.968	0.979	0.875	0.912
$O + C$	0.971	0.981	0.886	0.922
<i>All</i>	0.972	0.983	0.891	0.928

TABLE V
ABLATION STUDY FOR INTER-SUBBAND COLLABORATION MECHANISM

Methods	Celeb-DFv2	DFD	DFDC	FFIW
IE	0.968	0.980	0.878	0.916
CE w/o PE	0.970	0.982	0.887	0.924
<i>All</i>	0.972	0.983	0.891	0.928

C. Ablation Study

In this section, for evaluating the effectiveness of our designed Hierarchical Spatial-Frequency Fusion (HSFF), Collaborative Multi-Subband Enhancement (CMSE), we conduct detailed ablation studies through training on the FF++ and subsequent testing on Celeb-DFv2, DFD, DFDC, and FFIW.

For simplicity, Original backbone, HSFF, CMSE are denoted as O , H , C , respectively.

1) *Effectiveness of HSFF, CMSE*: As shown in Table IV, incorporating the HSFF module ($O + H$) yields consistent gains across all datasets, underscoring the importance of hierarchical frequency feature enhancement. To further justify the complexity of the multi-scale design, we conduct a fine-grained ablation study on the kernel sizes (from 3×3 to 9×9) within HSFF; the results (see Table V) show that each scale contributes positively,

and removing any single kernel leads to performance drops, confirming that the hierarchical structure is necessary. Interestingly, the full CMSE variant outperforms the simplified collaborative version, suggesting that inter-subband interaction can be more effective than enforcing independently enhancing subbands in certain complex scenarios. Nonetheless, the complete framework (*All*) achieves the best performance on all benchmarks, demonstrating that HSFF and CMSE are complementary and together enable more robust and generalizable deepfake detection.

2) *Effectiveness of Inter-Subband Collaboration Mechanism*: During the decomposition process, high-frequency subbands at different orientations and scales in DTCWT are not entirely independent, but exhibit structured information overlap and mutual reinforcement.

To further validate the contribution of the inter-subband collaboration mechanism in our proposed CMSE module, we conduct an ablation study with the HSFF module comparing different enhancement strategies, as summarized in Table V. Here, IE denotes individual enhancement applied separately to the six directional subbands of DTCWT, treating subbands as independent processing units, CE w/o PE refers to collaborative enhancement across subbands without phase enhancement, and *All* represents the full model incorporating both collaborative enhancement and phase enhancement.

The results demonstrate that CE w/o PE consistently outperforms IE across all four benchmark datasets. This improvement confirms that modeling inter-subband collaboration—pulling closer relevant subbands while pushing apart irrelevant ones—effectively enhances the discriminative capability of frequency representations.

Furthermore, the *All* configuration achieves the highest performance on all datasets, surpassing CE w/o PE by an additional margin. This indicates that the introduced phase enhancement (PE) further refines the complex frequency representations, complementing the inter-subband collaboration mechanism and contributing to more robust and generalizable detection.

Overall, the ablation study verifies that both collaborative enhancement and phase enhancement are essential components.

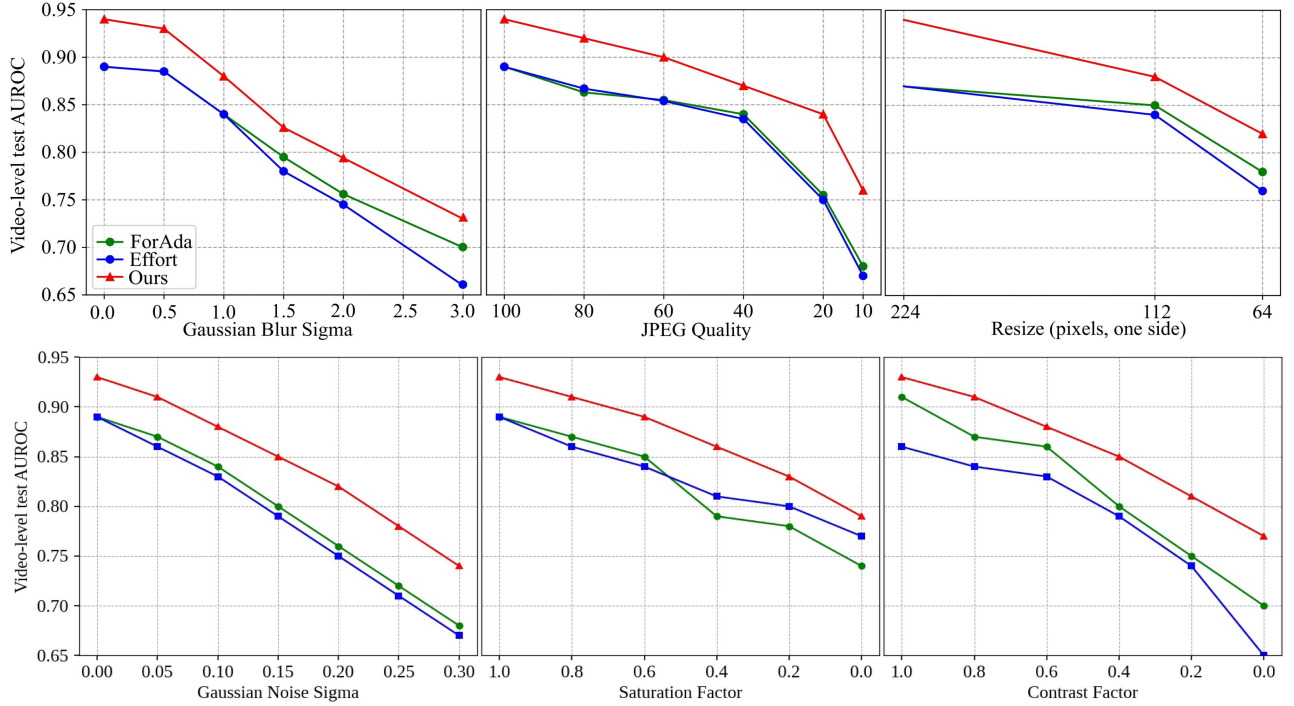


Fig. 5. Robustness comparison of deepfake detection methods under image degradations. Results are reported in terms of video-level AUROC (%) for Ours, ForAda [16], and Effort [17]. For resizing experiments, results are averaged over six interpolation strategies: nearest, Lanczos, bilinear, bicubic, box, and Hamming.

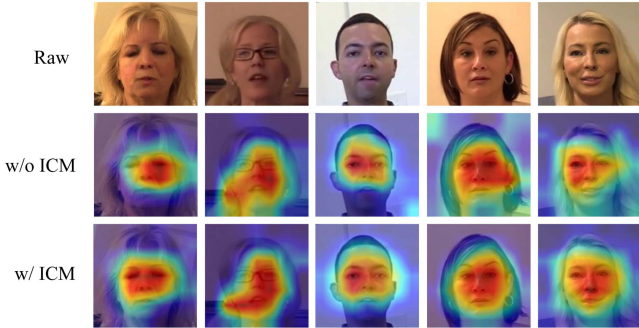


Fig. 6. Visualization of model's behaviour with or without the ICM by Grad-CAM.

Their coupling enables the model to better capture subtle, direction-sensitive forgery traces, leading to state-of-the-art cross-dataset generalization performance.

D. Visualisation Analysis

1) *Inter-Subband Collaboration Mechanism Analysis:* Inter-subband information collaboration represents pivotal aspect of our method. For CMSE, we perform both individual and collaborative enhancement on the six subbands, with corresponding visualizations provided. As illustrated in Fig. 6, our approach effectively identifies forged regions in manipulated images under both enhancement strategies. Nevertheless, CMSE with the ICM yields superior detection performance.

2) *Robustness Analysis:* We assess the robustness against commonly employed image degradation techniques, which are often utilized to erase manipulation watermarks and degrade the efficacy of deepfake detection. To ensure statistical reliability, we average the results of models trained with 5 different random seeds. The aggregated performance is illustrated in Fig. 5.

Six types of degradations were examined: Gaussian blurring, JPEG compression, image resizing, Gaussian Noise, Saturation, Contrast. For Gaussian blur, the sigma (σ) parameter was varied, with the corresponding kernel size set as $k = 2 \cdot \lceil 3\sigma \rceil + 1$, which enables the model to capture discriminative frequency-structured features consistently across multiple scales. In resizing experiments, facial images were downsampled to resolutions of 224×224 , 112×112 , and 64×64 pixels using six interpolation approaches: nearest-neighbor, Lanczos, bilinear, bicubic, box, and Hamming. Subsequently, each downsampled image was restored to the model's input resolution through the respective preprocessing pipeline of each method. For Gaussian noise, the standard deviation (σ) was varied from 0.00 to 0.30 with a step of 0.05. Reported results represent averages taken across all interpolation strategies.

V. CONCLUSION

In this paper, we propose a novel deepfake detection method that jointly harnesses the generalization capability of CLIP-ViT and the fine-grained frequency analysis via DTCWT. Our HSFF and CMSE modules aim to capture hierarchical frequency anomalies and enhance inter-subband correlations, respectively. This dual design thereby sharpens the model's discrimination

of spatial-frequency inconsistencies and bolsters its robustness against multi-directional forgery patterns, so as to mine fundamental implicit clues for robust discriminative features. Experimental results demonstrate state-of-the-art cross-dataset generalization performance on benchmarks including Celeb-DFv2, DFDC, FFIW, and DF40. Ablation studies and visualizations further validate the effectiveness of our approach.

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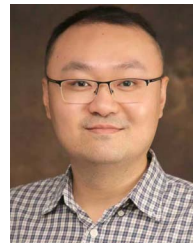
Rui Sun (Member, IEEE) received the PhD degree from the Huazhong University of Science and Technology of China, in 2003. He was a visiting scholar with Computer Science Department, University of Missouri-Columbia, USA from 2010 to 2011. He is currently a professor with the School of Computer Science and Information Engineering of Hefei University of Technology, China. His research interests include object recognition and tracking, computer vision, and machine learning.



Yifan Zhang received the BE degree from the Hefei University of Technology, in 2023. She is currently working toward the ME degree with the Hefei University of Technology, China. Her research interests include machine learning and computer vision.



Xiaolu Yu received the BE degree from Yanshan University, in 2022. He is currently working toward the ME degree with the Hefei University of Technology, China. His research interests include machine learning, computer vision. Currently engaged in research on deep forgery detection.



Meng Li (Senior Member, IEEE) received the PhD degree in computer science and technology from the School of Computer Science and Technology, Beijing Institute of Technology (BIT), China, in 2019. He is a professor with the School of Computer Science and Information Engineering, Hefei University of Technology (HFUT), China. He was a post-doc researcher with the Department of Mathematics and HIT Center, University of Padua, Italy, where he is with the Security and PRIVacy Through Zeal (SPRITZ) research group led by Prof. Mauro Conti (IEEE Fellow). He was sponsored by ERCIM 'Alain Bensoussan' Fellowship Programme (from 2020.10.1 to 2021.3.31) to conduct Post-Doc research supervised by Prof. Fabio Martinelli at CNR, Italy. He was sponsored by China Scholarship Council (CSC) as a Joint PhD student (from 2017.9.1 to 2018.8.31) supervised by Prof. Xiaodong Lin (IEEE Fellow) with the Broadband Communications Research (BBCR) Lab, University of Waterloo and Wilfrid Laurier University, Canada. He is supported by CSC as a visiting scholar (from 2025.3.1 to 2025.6.30) collaborating with Prof. Mauro Conti (IEEE Fellow) with the HIT Center, University of Padua, Italy. He was promoted to a professor from December, 2024. His research interests include secure and trustworthy data sharing, privacy preservation in IoV, applied cryptography, blockchain, TEE. In this area, he has published 165 papers in topmost journals and conferences, including *IEEE Transactions on Information Forensics and Security*, *IEEE Transactions on Dependable and Secure Computing*, *IEEE/ACM Transactions on Networking*, *IEEE Transactions on Mobile Computing*, *IEEE Journal on Selected Areas in Communications*, *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions on Technology and Society*, *IEEE Transactions on Parallel and Distributed Systems*, *IEEE Transactions on Software Engineering*, *IEEE Transactions on Services Computing*, COMST, IEEE S&P, USENIX Security, MobiCom, and INFOCOM. He is a senior member of CCF, CACR, CIE, and CIC. He is a senior area editor for *IEEE Transactions on Information Forensics and Security* and an associate editor for *IEEE Transactions on Dependable and Secure Computing*. He has served as a TPC member for conferences, including ICDCS, Inscrypt, ACISP, ICICS, and TrustCom. He is the recipient of 2024 IEEE HITC Award for Excellence (Early Career Researcher) and 2025 IEEE TCSVC Rising Star Award. He was selected into the IEEE Computer Society Computing's Top 30 Early Career Professionals for 2025.



Jun Gao received the bachelor's degree in electronic engineering and the master's degree in signal and information processing from the Hefei University of Technology, in 1985 and 1991, respectively, and the PhD degree from the Hefei University of Science and Technology, in 1999. He is currently a professor with the Laboratory of Image Information Processing, HFUT. His current research interests include image processing and intelligent information processing.